THE LOCUS OF THE BASE WHERE A CLOSED MAP BECOMES UNRAMIFIED IS AN OPEN SUBSPACE WITH THE UNIVERSAL PROPERTY

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 11528-11541 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 11543-11563, the complete author proof (19 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Spaces avatar of the schemes-side where-closed-immersion-open family (L16084 collected); this one is the where-unramified descent with a genuinely different construction.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Morphisms of Spaces, lemma where-unramified (the unramified-locus open and its base-change compatibility); Properties of Spaces, lemmas open-subspaces and factor-through-open-subspace (the dictionary between open subspaces and their point sets, and the factoring criterion).

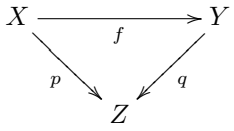
Source reference. Original lemma 05W2: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 11563–11563. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03IE: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 11563–11563. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03BZ: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 11563–11563. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Consider a commutative diagram*



of algebraic spaces. Assume that p is locally of finite type and closed. Then there exists an open subspace $W \subset Z$ such that a morphism $Z' \to Z$ factors through W if and only if the base change $f_{Z'} : X_{Z'} \to Y_{Z'}$ is unramified.

Proof. By Morphisms of Spaces, Lemma 05W2 there exists an open subspace $U(f) \subset X$ which is the set of points where f is unramified. Moreover, formation of $U(f)$ commutes with arbitrary base change. Let $W \subset Z$ be the open subspace (see Properties of Spaces, Lemma 03BZ) with underlying set of points

$$|W| = |Z| \setminus |p| (|X| \setminus |U(f)|)$$

i.e., $z \in |Z|$ is a point of W if and only if f is unramified at every point of X above z. Note that this is open because we assumed that p is closed. Since the formation of $U(f)$ commutes with arbitrary base change we immediately see (using Properties of Spaces, Lemma 03IE) that W has the desired universal property. □